

# Ivan Sysoev

Play Technology Developer

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Portfolio: <https://isysoev.github.io>

Tingstedet 56, Billund, Denmark, 7190

Experienced developer and designer of play technologies with strong technical background and experience of bridging technical and non-technical domains. Interested in creating innovative learning experiences that cultivate children's curiosity, creativity and sense of empowerment.

## EXPERIENCE

*Creative Play Lab, LEGO Group*

Billund, Denmark

Senior Staff Engineer

2023 – present

- Developing advanced prototypes to support play exploration.
- Informing tech development with lessons from play.

*MIT Media Lab*

Cambridge, MA

Postdoctoral Associate

2020 – 2023

- Designed and implemented robotic companions to support development of early literacy and creativity in children via open-ended play.
- Designed and implemented an interactive storybook to introduce preschoolers to artificial intelligence concepts.

*MIT Media Lab*

Cambridge, MA

Research Assistant

2014 – 2020

- Proposed a child-driven, machine-guided approach to early literacy learning, grounded in findings from learning theory and literacy learning research.
- Designed and developed two open-ended literacy apps to evaluate this approach.
- Refined the proposed designs via iterative playtesting.
- Conducted mixed-method studies utilizing Design-Based Research methodology to evaluate the approach.
- Closely collaborated with colleagues and schools to conduct the studies.

*Ledas Ltd. / Bricsys Technologies Russia*

Novosibirsk, Russia

Software Developer

2009 – 2012

- Researched a novel approach to modeling in CAD systems
- Developed CAD software on C++
- Participated in designing the architecture of the system

## EDUCATION

**Massachusetts Institute of Technology**

Cambridge, MA

PhD in Media Arts and Sciences, GPA: 5.0 / 5

2014 – 2020

**Georgia Institute of Technology**

Atlanta, GA

Master of Science in Computer Science, GPA: 4.0 / 4

2012 – 2014

**Novosibirsk State University**

Novosibirsk, Russia

Master of Science in Computer Science, GPA: 4.86 / 5

2009 – 2011

**Novosibirsk State University**

Novosibirsk, Russia

Bachelor of Science in Computer Science, GPA: 5.0 / 5

2005 – 2009

## RESEARCH PROJECTS HIGHLIGHT

A constructionist approach to early literacy

MIT Media Lab, 2015-2018

- Based on the literature on constructionism, Montessori education and emergent literacy, designed and implemented an early literacy app *SpeechBlocks*.
- Led a preschool-based pilot study with *SpeechBlocks* and participated in two home-based studies with the app led by my colleagues.
- Key findings:
  - The approach facilitates children's engagement, agency and self-efficacy;
  - Social interactions around literacy activities emerged between children;
  - There is a need for automatic scaffolding (guidance) for the approach to be scalable.
- Related publication: (3)

Child-driven, machine-guided learning

MIT Media Lab, 2017-present

- Based on results of the previous project, proposed a child-driven, machine-guided approach and implemented an app *SpeechBlocks II* to evaluate it.
- Implemented a novel information-theoretic method to align phonemes and graphemes in English words, to support the work of the scaffolding system.
- Led a school-based study with 4-5 years old participants to evaluate the approach.
- Key findings:
  - Most children were eventually able to use the system nearly autonomously, which supported their expressive play and peer learning;
  - Children with lower self-regulation and literacy skills were more likely to engage in distracted behaviors, which reduced the effectiveness of the system for them.
- Related publication: (1)

Input mechanisms for child-driven literacy learning

MIT Media Lab, 2018-2019

- Implemented input mechanisms for children to communicate their intent to *SpeechBlocks II*, to support child-driven literacy learning activities: word bank, speech recognition, text recognition, invented spelling interpretation, semantic association network (the last four were innovative).
- Evaluated the design in the *SpeechBlocks II* school-based study with 4-5 year olds.
- Key findings:
  - Word bank, speech recognition, text recognition and the association network were actively used;
  - Three roles of different input mechanisms emerged: (1) helping the child implement specific

- ideas, (2) helping the child to browse for ideas, and (3) being a fall-back option;
- Invented spelling interpretation was difficult to use, and text recognition led to frequent distractions.

#### Phoneme-based blocks for early literacy apps

MIT Media Lab, 2018-2019

- Suggested using phoneme blocks to avoid the issue of orthographic complexity, which is known to interfere with early literacy learning.
- Determined optimal block design via iterative playtesting. Designed and implemented onomatopoeic characters to represent phonemes.
- Evaluated the design in the *SpeechBlocks II* school-based study with 4-5 year olds.
- Key findings:
  - Children generally found the characters engaging and understood their functioning;
  - There were differences in effectiveness of onomatopoeic mnemonics for different children, possibly determined by their preexisting letter-sound knowledge.
- Related publication: (2)

#### SELECT PUBLICATIONS

(1) **Sysoev, I.**, Gray, J. H., Fine, S., Makini, S.P. & Roy, D. (2022). Child-driven, machine-guided: Automatic scaffolding of constructionist-inspired early literacy play. *Computers & Education*.

(2) **Sysoev, I.**, Gray, J. H., Fine, S., & Roy, D. (2021). Designing building blocks for open-ended early literacy software. *International Journal of Child-Computer Interaction*.

(3) **Sysoev, I.**, Hershman, A., Fine, S., Traweek, C., & Roy, D. (2017). SpeechBlocks: A Constructionist Early Literacy App. *Proceedings of the 2017 Conference on Interaction Design and Children*

#### SKILLS

- Designing learning technologies for children
- Qualitative and quantitative research
- Programming languages: C++, C#, Java, JavaScript, Python, Lisp, Haskell, MATLAB
- Object-oriented design and functional programming
- Game development in Unity
- Android development
- Machine learning, AI and NLP
- Oral and written proficiency in Russian.
- Drawing and painting using traditional and digital media; digital animation

#### AWARDS

- Fulbright Visiting Graduate Student Scholarship 2012 – 2014
- Donald Jackson Fellowship 2013  
*Awarded to 3 Georgia Tech College of Computing MS students yearly*
- Baker Atlas Fellowship 2009  
*Awarded to students and alumni of Novosibirsk State University*

## COMMUNITY INVOLVEMENT

*MIT*

2016-2019

Recycling and Gardening Chair at a graduate dorm; Sustainability Subcommittee Member

*826 Boston*

2016-2018

Volunteer in an afterschool program.